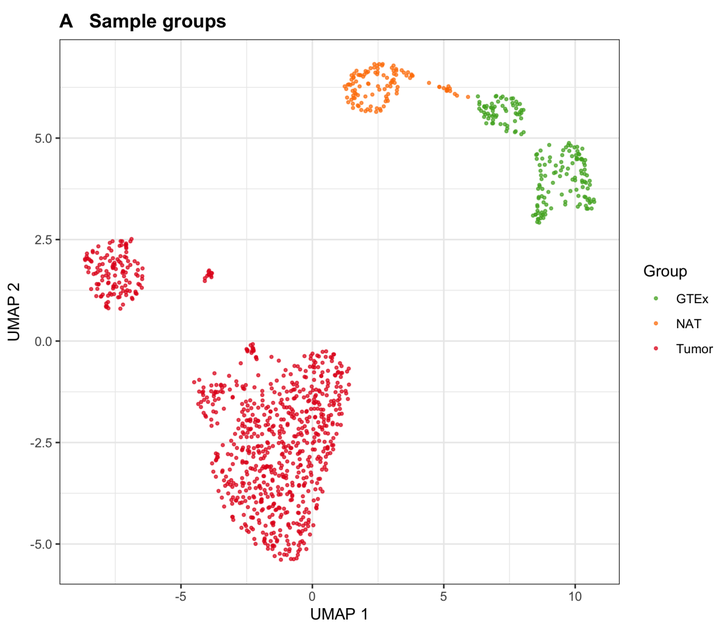
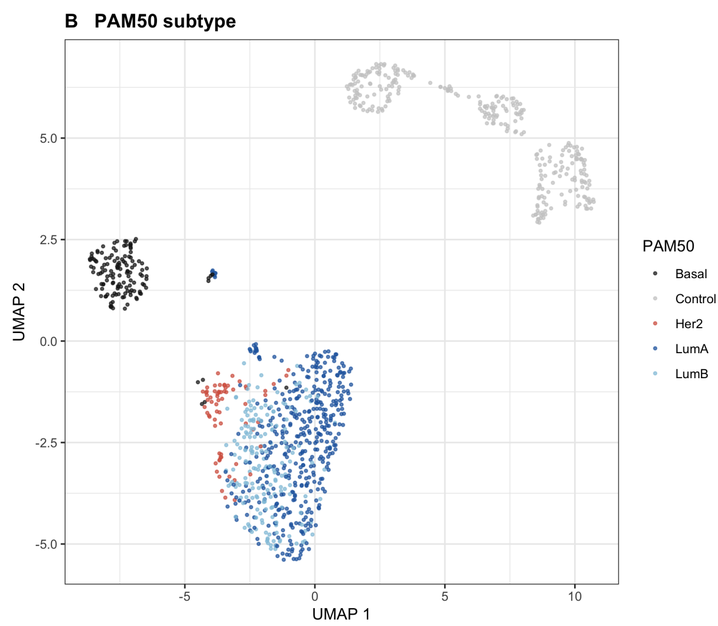
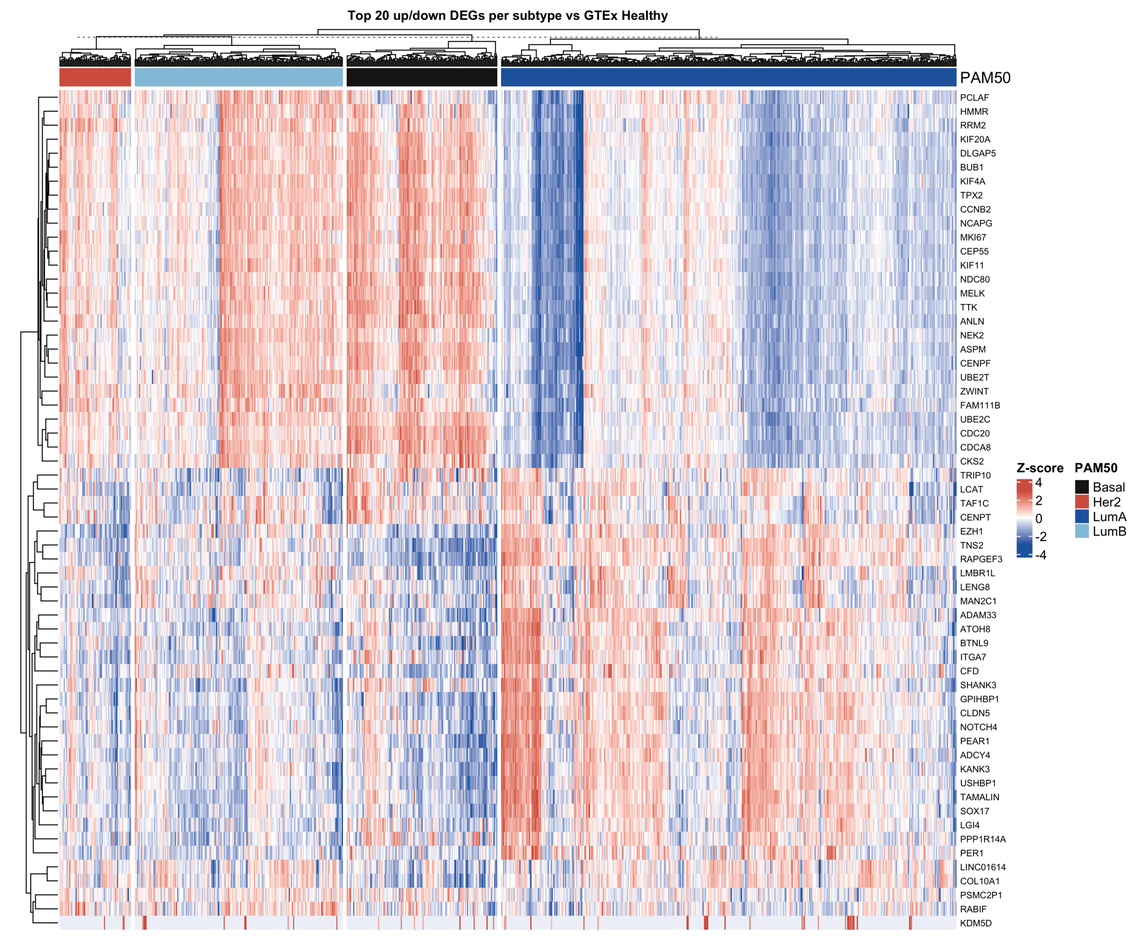
TCGA−BRCA PAM50 Subtype Analysis − Pipeline Summary

A UMAP − sample groups and PAM50 subtypes

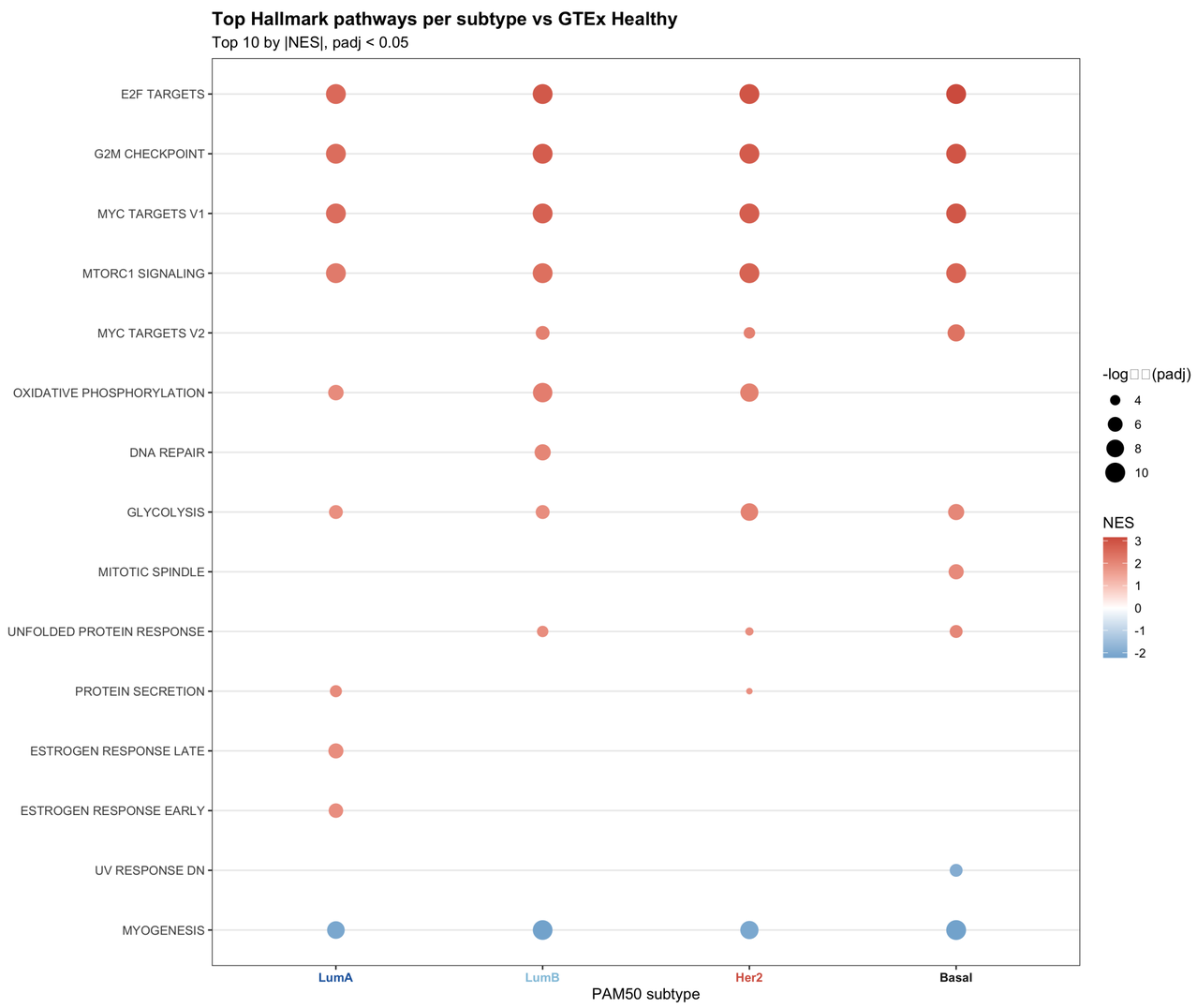




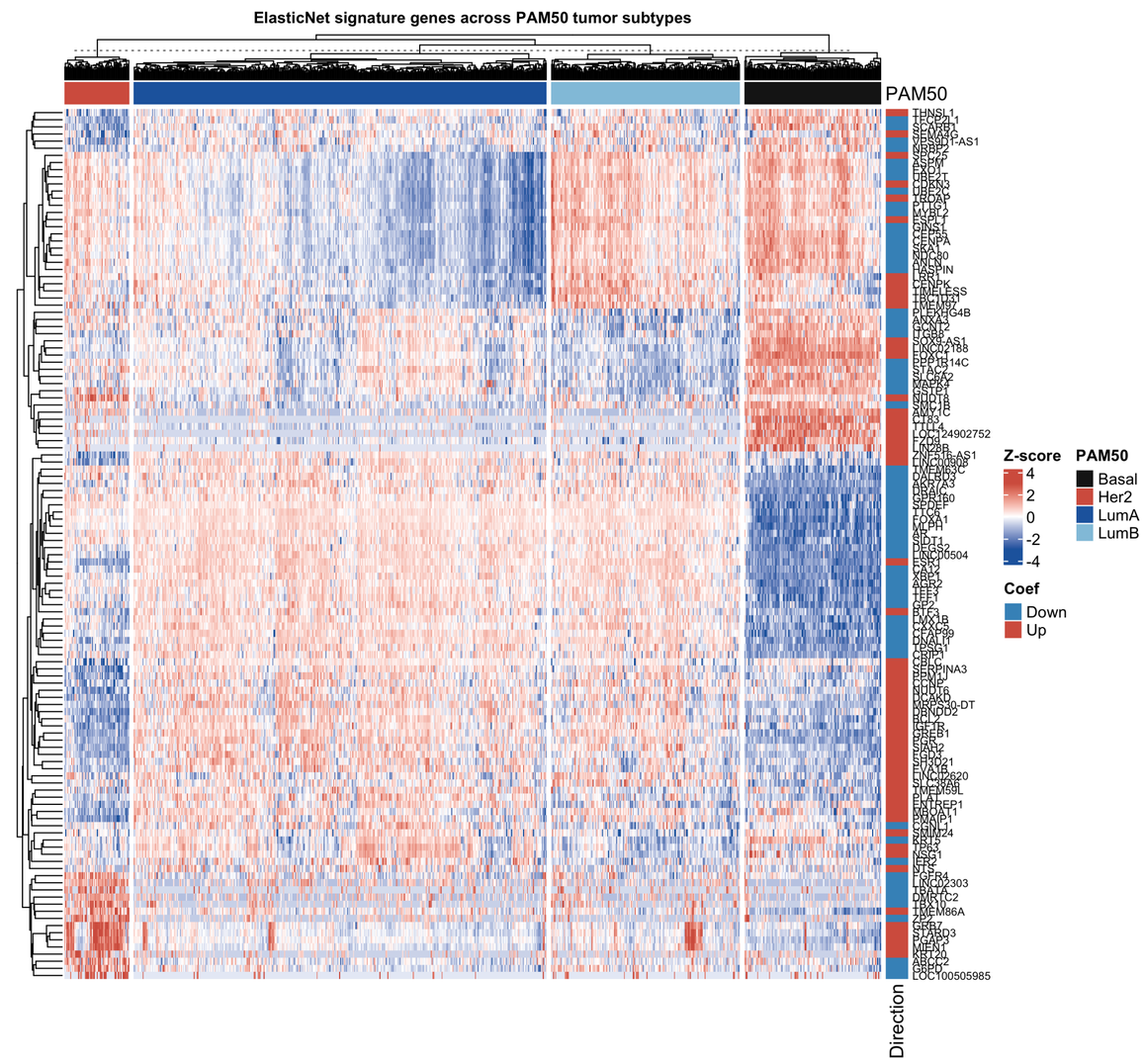
B Top DEGs per subtype vs GTEx (z−scored)



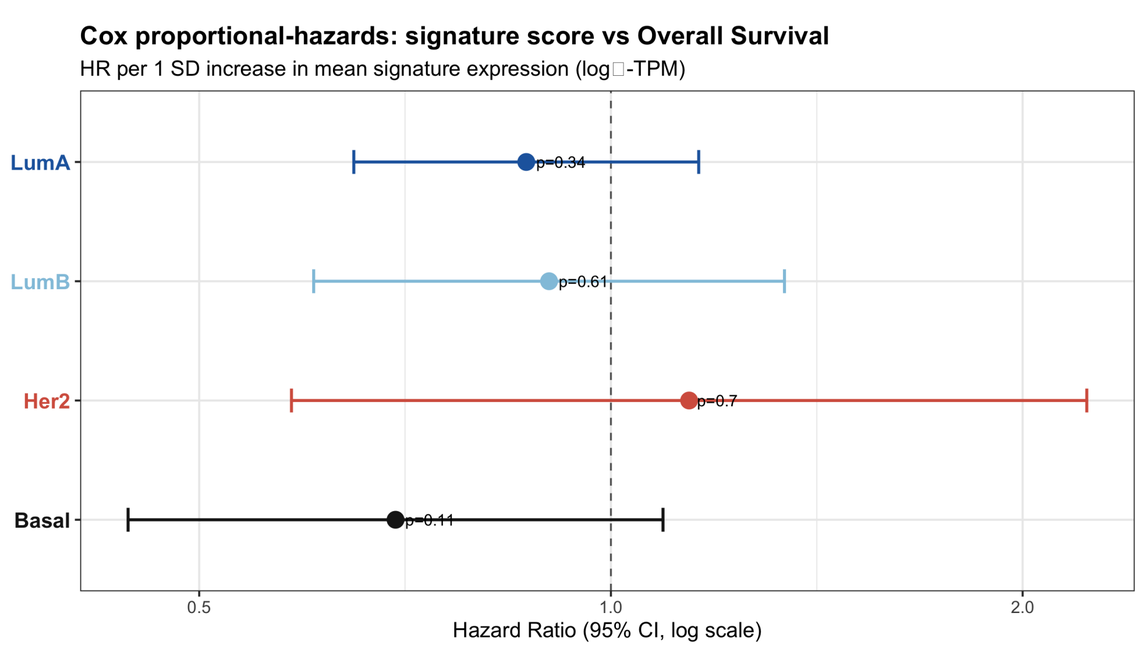
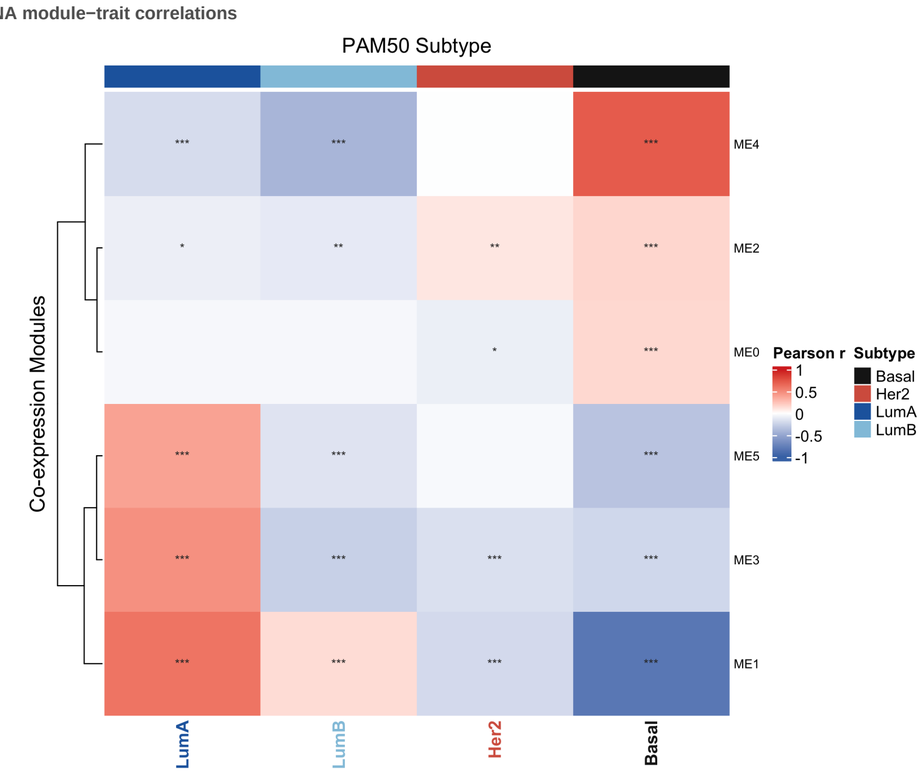
C Hallmark pathway enrichment (fgsea NES)



D ElasticNet signature genes across tumors



E Cox PH: signature score vs Overall Survival



F WGCNA module−trait correlations